

May/June 2013

Time : 2 hours

Candidates may attempt ALL questions in Section A and at most TWO questions in Section B. Each question should start on a fresh page.

SECTION A (40 marks)

Candidates may attempt ALL questions being careful to number them A1 to A5.

A1. Define what is meant by saying that

(a) a function $f(x, y)$ has a **limit**, L , at a point (a, b) as (x, y) approaches (a, b) . [2]

(b) a function $f(x, y)$ has a **partial derivative** $f_x(a, b)$ with respect to x at a point (a, b) . [2]

(c) the series $\sum_{n=1}^{\infty} a_n$ is **absolutely convergent**. [2]

(d) the series $\sum_{n=1}^{\infty} a_n$ is **conditionally convergent**. [2]

A2. (a) Find the area of the triangle with vertices $(-1, 2, 3)$, $(2, -3, 1)$ and $(1, -1, 2)$. [4]

(b) Find the area of the parallelogram determined by $\mathbf{a} = 3\mathbf{i} + 4\mathbf{j}$ and $\mathbf{b} = \mathbf{i} + \mathbf{j} + \mathbf{k}$. [4]

A3. Let $f(x, y)$ be a function of x and y where $x = e^{u-v} \cos(u+v)$ and $y = e^{u-v} \sin(u+v)$. Show that

$$\frac{\partial f}{\partial u} + \frac{\partial f}{\partial v} = 2x \frac{\partial f}{\partial y} - 2y \frac{\partial f}{\partial x}.$$

[6]

A4. Let

$$f(x, y) = \begin{cases} \frac{xy}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$$

- (a) Determine whether or not $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ exists at the origin, and find their values if they exist. [4]
- (b) Does $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ exist? [4]

A5. (a) Determine whether or not the series converges and find its sum if it converges

$$\sum_{r=1}^{\infty} \frac{20}{(7r-3)(7r+4)} = \frac{20}{4 \cdot 11} + \frac{20}{11 \cdot 18} + \frac{20}{18 \cdot 25} + \cdots + \frac{20}{(7n-3)(7n+4)} + \cdots$$

[5]

- (b) By interchanging the order of integration or otherwise, evaluate the integral

$$\int_0^1 \int_{3y}^3 e^{x^2} dx dy.$$

[5]

SECTION B (60 marks)

Candidates may attempt TWO questions being careful to number them B6 to B9.

- B6.** (a) (i) Write down the equation of the tangent plane to the ellipsoid $2x^2 + 4y^2 + z^2 - 45 = 0$ at the point $(2, -3, -1)$. [5]
- (ii) Find the equation of the plane passing through the points $(2, -1, 2)$, $(-1, 2, -3)$ and $(4, 1, 0)$. [5]
- (b) (i) Find the angle between the vectors $\mathbf{a} = 2\mathbf{i} + 4\mathbf{j} + 6\mathbf{k}$ and $\mathbf{b} = \mathbf{i} - 3\mathbf{j} + 2\mathbf{k}$. [4]
- (ii) Find the parametric equations of the line of intersection of the planes

$$2x + 3y - z = -3 \quad \text{and} \quad 4x + 5y + z = 1$$

and calculate the angle between them. [8]

- (c) (i) Find the parametric and symmetric equations for the straight line through $(2, 5, -7)$ and $(4, 3, 8)$. [4]
- (ii) Find the distance between the point $(5, 12, -13)$ and the plane $3x + 4y + 5z = 12$. [4]

- B7.** (a) Using the definition of a limit, show that $\lim_{(x,y) \rightarrow (2,1)} (xy - 3x + 4) = 0$. [6]
- (b) Find and classify the stationary points of the function

$$f(x, y) = x^3 + y^3 - 3x - 12y + 20.$$

[6]

- (c) Parabolic co-ordinates (u, v) are defined implicitly in terms of the Cartesian co-ordinates (x, y) by the pair of equations

$$x = \frac{u^2 - v^2}{2} \quad \text{and} \quad y = uv.$$

- (i) Obtain expressions for $\frac{\partial u}{\partial x}$, $\frac{\partial u}{\partial y}$, $\frac{\partial v}{\partial x}$ and $\frac{\partial v}{\partial y}$. [8]
- (ii) Given that $f(x, y) = \phi(u, v)$. Obtain expressions for $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ in terms of $\frac{\partial \phi}{\partial u}$, $\frac{\partial \phi}{\partial v}$, u and v and deduce that

$$\left(\frac{\partial f}{\partial x}\right)^2 + \left(\frac{\partial f}{\partial y}\right)^2 = \frac{1}{u^2 + v^2} \left[\left(\frac{\partial \phi}{\partial u}\right)^2 + \left(\frac{\partial \phi}{\partial v}\right)^2 \right].$$

[10]

- B8.** (a) Sketch the region of integration and evaluate the following integrals.

(i) $\int_0^1 \int_y^{\sqrt{y}} dx dy.$

(ii) $\iint_D \sqrt{x^2 + y^2} dy dx$ where D is the region bounded by $x^2 + y^2 \geq 4$ and $x^2 + y^2 \leq 9$. [3,3]

(b) (i) Evaluate $\int_0^1 \int_0^1 \int_{\sqrt{x^2+y^2}}^2 xyz dz dy dx.$ [5]

- (ii) Evaluate the following integral, using an appropriate transformation or otherwise

$$\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{dx dy}{1 + x^2 + y^2}.$$

[5]

- (c) Let $u = x^2 - y^2$ and $v = 2xy$.

(i) Show that $(x^2 + y^2)^2 = u^2 + v^2$. [3]

(ii) Hence, find $\frac{\partial(x, y)}{\partial(u, v)}$ in terms of u and v . [5]

(iii) Hence, evaluate $\iint_R (x^2 + y^2) dx dy$ where R is the region in the first quadrant of the xy -plane bounded by the curves $x^2 - y^2 = 1$, $x^2 - y^2 = 9$, $xy = 2$ and $xy = 4$. [6]

B9. (a) Test for convergence or divergence for each of the following infinite series by applying an appropriate test.

$$(i) \sum_{n=1}^{\infty} \frac{n}{2^n} \quad (ii) \sum_{n=1}^{\infty} \frac{n-7}{4n^3+3} \quad (iii) \sum_{n=1}^{\infty} \frac{n}{\sqrt{n^4+1}}. \quad [3,3,3]$$

$$(b) (i) \text{ Find the sum of the series } \sum_{n=1}^{\infty} \frac{1}{n(n+1)}. \quad [4]$$

(ii) Consider the series $\sum_{n=1}^{\infty} ar^{n-1}$ where a and r are constants. Determine when the series converges and when it diverges and find the sum when it converges. [5]

(c) Test the following series for absolute and conditional convergence and determine the interval of convergence.

$$(i) \sum_{n=1}^{\infty} \frac{(-1)^n x^n}{n2^n}.$$

$$(ii) \sum_{n=1}^{\infty} \frac{n^3 x^n}{n!}.$$

[6,6]

END OF QUESTION PAPER